

Business Analytics Case Study Report: Telecom Customer Churn Analysis

1. Introduction

Customer retention is a critical success factor in the telecommunications industry because revenue depends heavily on long-term customer relationships. Customer churn occurs when subscribers discontinue their services, reducing recurring revenue and increasing acquisition costs. This report analyses customer churn patterns using business analytics techniques to identify key churn drivers and recommend strategies to improve retention.

2. Business Problem Statement

The organisation seeks to understand why customers churn, which customers are most likely to churn, and which controllable business factors contribute most to customer attrition. The goal is to generate evidence-based recommendations that improve customer retention and long-term profitability.

3. Dataset Description

The dataset contains 7,043 customer records and 21 variables covering customer demographics, subscribed services, billing information, payment methods and churn status. Variables include gender, senior citizen status, partner, dependents, tenure, phone service, internet service, online services, technical support, contract type, payment method, monthly charges, total charges and churn.

4. Data Analysis Process

Data preparation involved standardising headers, checking duplicates, reviewing missing Total Charges values, validating categorical values and generating descriptive statistics. Microsoft Excel PivotTables and PivotCharts were used to analyse churn across demographic, service, billing and tenure dimensions. Interactive dashboards were then developed for Executive Overview, Customer Profile, Services, Billing and Executive Insights.

5. Key Findings

- Overall churn rate was 26.5% (1,869 of 7,043 customers).
- Month-to-month contracts recorded the highest churn.
- High monthly charges were associated with higher churn, particularly for month-to-month customers.
- Electronic Check customers exhibited the highest churn among payment methods.
- Customers without Technical Support and other value-added services churned more frequently.
- New and early-tenure customers were the most vulnerable to churn.

6. Business Insights

The analysis shows that churn is driven primarily by controllable business factors rather than customer demographics. Pricing, contract type, payment method, tenure and service adoption collectively define the highest-risk customer segments. These insights provide management with clear opportunities to improve customer retention through targeted interventions.

7. Recommendations

1. Encourage migration from month-to-month to longer-term contracts.
2. Review pricing for high monthly charge plans.
3. Improve customer onboarding during the first year.
4. Promote Technical Support, Online Security and Device Protection services.
5. Encourage automatic payment methods over Electronic Check.
6. Implement a proactive churn monitoring and retention programme.

8. Conclusion

The project demonstrates how business analytics converts operational data into actionable business intelligence. The dashboard provides decision-makers with clear visibility into churn drivers and supports informed strategies to improve customer retention, customer lifetime value and sustainable business growth.